

Initial Project Planning Template

Date:	01.10.2026
Team ID:	6ab295b91266c775575a6dce
Project Name:	CRM APPLICATION FOR THE CHAI SHOP & CO
Maximum Marks:	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Registration	USN-1	As a user, I can register by entering mobile number & OTP verification for Chai Shop CRM	2	High	Sivaranjani E	15 Sep 2026	16 Sep 2026
Sprint-1	Registration	USN-2	As a user, I will receive confirmation SMS once registered for Chai Shop CRM	1	High	Snega B	15 Sep 2026	16 Sep 2026
Sprint-2	Customer Mgmt	USN-3	As an owner, I can add/view/edit/delete customer profiles with purchase history	2	Low	Sudhesh P	17 Sep 2026	18 Sep 2026
Sprint-1	Billing	USN-4	As a staff, I can create fast billing under 2 sec with auto inventory deduction	2	Medium	Sivashakthi S	17 Sep 2026	18 Sep 2026
Sprint-1	Login	USN-5	As a user, I can log into app by entering mobile number & OTP and role selection Owner/Staff/Customer	1	High	Suganya E	17 Sep 2026	18 Sep 2026

Team Size: 5 | Sivaranjani E, Sivashakthi S, Snega B, Sudhesh P, Suganya E | Team ID: 6ab295b91266c775575a6dce | Date: 01.10.2026

Code-Layout, Readability and Reusability

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Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development. For CRM Chai Shop we followed PEP8, MVC pattern and component reusability.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP8 followed, 4-space indentation in Python/Flask &
2	Proper File Structure	Files and folders are logically organized	Yes	MVC: /models /controllers /routes /views /services for CRM
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	custName, chaiStock, loyaltyPoints, orderTotal - self
4	Function / Method Names	Functions are descriptively named	Yes	addCustomer(), generateBill(), calculateLoyalty(),
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings for all APIs, comments for billing & Razorpay
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for Customer, Billing, Loyalty, Inventory,
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common utils reused, no duplicate OTP/billing logic
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-catch for OTP, Payment failure, Inventory <5kg alert

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Customer Management Module	Python / Flask, React	Add/Edit/Delete/View customer profiles, loyalty history	High
2	Billing Engine	Python / FastAPI, JS	Fast billing <2 sec, auto inventory deduction, PDF gen	High
3	Loyalty Program Service	Python	Auto add 1 point per Rs10, 10% off on 5th order calculation	High
4	Inventory Manager	Python / MySQL	Chai stock tracking, alert when stock below 5kg	Medium
5	Payment & Notification Service	Razorpay, Fast2SMS, WhatsApp Business API	UPI/Cash payment, order ready & offer alerts	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Structure & Layout	5	Well structured MVC, separate folders for CRM modules, clean file organization
Readability	5	Meaningful names, proper indentation, docstrings, easy for team of 5 to understand
Reusability	5	Billing, Loyalty, Notification modules highly reusable for future Chai Shop branches
Maintainability	4	Error handling for OTP/Payment, easy to extend for new offers & inventory

Coding & Solution

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Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Sivaranjani-E/CRM-Chai-Shop-Co Team: 6ab295b91266c775575a6dce All source code, .env.example, requirements.txt included
Programming Language(s)	Python, JavaScript
Framework(s) Used	Flask, React JS, MySQL, Razorpay SDK, Fast2SMS, WhatsApp Business API
Key Features Implemented	OTP Registration/Login, Customer CRUD, Sales Dashboard, Fast Billing <2sec auto inventory, Bill PDF via WhatsApp/SMS, Loyalty 1 point/Rs10 & 10% off 5th order, Stock alert <5kg, Razorpay UPI/Cash, Feedback & Referral
Pending / Incomplete Features	None - All implemented Future: Multi-branch, AI sales prediction
Setup / Run Instructions	Clone repo > pip install -r requirements.txt > npm install > Setup .env (DB, Razorpay, Fast2SMS, WA API) > python app.py > npm start Tested Python 3.10+ Team: Sivaranjani E, Sivashakthi S, Snega B, Sudhesh P, Suganya E

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes - Customer, Billing, Loyalty,
2	Meaningful variable and function names are used	Yes - custName, chaiStockKg,
3	Code includes comments / documentation where necessary	Yes - Docstrings for all APIs,
4	Error handling is implemented for critical operations	Yes - Try-catch for OTP
5	The application runs without critical errors	Yes - Tested fast billing <2 sec,
6	Code is committed to a version control repository	Yes - GitHub: CRM-Chai-Shop-Co

Additional Notes / Comments:

CRM Application for The Chai Shop & Co successfully implemented with all key features: Mobile OTP Registration/Login, Customer Management CRUD, Fast Billing <2 sec with auto inventory deduction, Sales Dashboard, Loyalty Program (1 point per Rs10, 10% off on 5th order), Inventory alert when stock below 5kg, Razorpay UPI/Cash payment, Bill PDF via WhatsApp/SMS using Fast2SMS & WhatsApp Business API, Feedback & Referral Rewards. Code follows PEP8, MVC pattern, modular reusable services for future branches. Team Size 5: Sivaranjani E, Sivashakthi S, Snega B, Sudhesh P, Suganya E. Team ID: 6ab295b91266c775575a6dce. Date:

No. of Functional Features Included in the Solution

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Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress)	Points Contribution
1	Customer Registration & OTP Login	Mobile number OTP verification, role selection	Auth Module	Done	1
2	Customer Management CRUD	Add/view/edit/delete customer profiles with purchase history &	Customer Module	Done	0.5
3	Sales Dashboard & Analytics	Real-time sales, top customers, daily revenue charts for	Dashboard Module	Done	0.5
4	Fast Billing System	Billing <2 sec, auto inventory deduction, chai stock update,	Billing Module	Done	1
5	Loyalty Program	Auto 1 point per Rs10, 10% off on 5th order, referral rewards auto	Loyalty Module	Done	0.5
6	Inventory Management	Chai stock tracking, alert when stock below 5kg, supplier	Inventory Module	Done	0.5
7	Payment Integration	Razorpay UPI/Cash payment, secure transaction, receipt	Payment Module	Done	0.5
8	Notification Service	Fast2SMS & WhatsApp Business API for order ready, offers, bill	Notification Module	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8 (100% - All Done)
Core / Must-Have Features	6 (Registration, Customer Mgmt, Billing, Inventory, Payment, Dashboard)
Additional / Nice-to-Have Features	2 (Loyalty Program, Notification Service)
Features Tested & Verified	8 - All tested: OTP login, Billing <2sec, Loyalty calc, Stock alert <5kg, Razorpay, WhatsApp/SMS

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Registration Login UI, Billing UI <2sec, Dashboard with charts - React JS
2	Backend / Logic	3	Customer CRUD, Loyalty calc 1 point/Rs10 & 10% off 5th order, Fast Billing logic - Flask/FastAPI
3	Database / Storage	2	Customer profiles, Inventory stock tracking MySQL - Chai stock alert <5kg
4	API / Integration	3	Razorpay UPI/Cash, Fast2SMS, WhatsApp Business API for bill PDF & alerts
5	Security / Authentication	1	Mobile OTP verification, hashed OTP, input validation, secure .env keys